BIRZEIT UNIVERSITY

COMP322: Human Computer Interaction (HCI)

Second Semester

Course Outline

Instructor:	Dr Hisham Ihshaish
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Course details

Textbooks

Primary (T1): Rogers, Y., Sharp, H., & Preece, J. (2024). *Interaction Design: Beyond Human-Computer Interaction* (6th ed.). Wiley. ISBN: 978-1119901099.

The 4th and 5th editions remain acceptable for most chapters. The 6th edition has substantial updates to Chapter 6 (Interfaces), adding coverage of voice, AI, and conversational interfaces. Companion website: id-book.com.

Secondary (T2): Lazar, J., Feng, J.H., & Hochheiser, H. (2017). *Research Methods in Human-Computer Interaction* (2nd ed.). Morgan Kaufmann.

References

- The Design of Everyday Things, Donald Norman (Revised Edition, 2013)
- Nielsen Norman Group: nngroup.com
- ACM Digital Library proceedings (CHI, CSCW, UIST)
- ISO 9241 standards series

HCI overview

Human-computer interaction (HCI) involves the study, planning, design, and uses of the interaction between people (users) and computers. It is the intersection of computer science, behavioural sciences, design, and human factors engineering. Attention to the design of human-machine interaction is important because poorly designed interfaces can lead to wasted time, frustration, exclusion of users with disabilities, and in some cases, serious harm.

HCI aims to improve interactions between users and computers by making systems more usable and responsive to users' needs. Specifically, HCI has interests in:

- Methodologies and processes for designing interfaces, optimising for desired properties such as learnability or efficiency of use
- Methods for implementing interfaces
- Techniques for evaluating and comparing interfaces
- Developing new interfaces and interaction techniques, including voice, AI-driven, and multimodal interaction
- Developing descriptive and predictive models and theories of interaction

Course objectives

By the end of this course, students should be able to:

- Apply the principles of HCI to the analysis and design of usable interfaces
- Work through the process of understanding users, specifying requirements, prototyping, and evaluating interactive systems
- Conduct effective usability evaluations using both inspection methods (heuristic evaluation) and empirical methods (usability testing)
- Design for accessibility and cultural context, including right-to-left layouts and Arabic-speaking users
- Critically engage with emerging interaction paradigms including AI and multimodal interfaces
- Work effectively in a team environment on a real design project

Evaluation

Component	Weight
Individual assignment	15%
In-class quiz	10%
Multi-phase group project	30%
Final presentation (project defence)	15%
Final exam	30%

Full details of the project, assignments, and presentation are provided in a separate document: COMP322 Project and Assignments Specification.

Course outline

Sessions run Saturday, Monday, and Wednesday, 50 minutes each.

Block	Sessions	Topic	Reading
0		Introduction to user-centred design and usability engineering	External material
1	3	What is interaction design?	T1 Ch 1
2	3	Understanding and conceptualising interaction	T1 Ch 2
3	3	Cognitive aspects	T1 Ch 3
4	3	Social and emotional interaction	T1 Ch 4 and 5
5	5	Interfaces (GUI, voice, gesture, AI, AR/VR, tangible)	T1 Ch 6
6	2	Data gathering	T1 Ch 7
7	3	Process, requirements, and prototyping	T1 Ch 9-11
8	2	Experimental research and design	T2 Ch 2-3
9	2	Data analysis, interpretation, and statistical analysis	T1 Ch 8; T2 Ch 4
10	3	Student project presentations	–

Block 0 will cover Norman's design principles, the UCD process (ISO 9241-210), usability goals (ISO 9241-11), Nielsen's 10 usability heuristics, the heuristic evaluation method, personas and scenarios, and accessibility foundations (WCAG 2.2 and universal design).

Special regulations

- Late assignments will not be accepted for any reason.
- There will be no make-up quizzes.
- Missing any exam without an acceptable excuse will result in a zero grade for that exam.
- Attendance is mandatory. University regulations will be strictly enforced.

Enjoy HCI!